

The Complete Blood Work Guide

What to check, what it tells you & how to understand your results

How to Use This Guide

Welcome to **The Complete Blood Work Guide**. This document is designed to give you a practical, easy-to-understand checklist of key blood markers to reference when discussing blood work with your healthcare professional.

Routine blood work provides an objective look under the hood, allowing you to monitor your metabolic function, hormonal balance, and organ health over time.

IMPORTANT DISCLAIMER: This guide is intended for educational purposes only and does not constitute medical advice. It is not a universal medical protocol. Not everyone needs every test listed here, nor is it necessary to test at the same frequency. Always consult with a qualified healthcare professional before making any decisions regarding your health.

01

CBC / Complete Blood Count

Red Blood Cells (RBC) 4.5 - 5.9 x10¹²/L

WHAT IT MEASURES

The number of red blood cells in your body.

WHY IT MAY MATTER

Low levels indicate anemia; high levels can point to polycythemia.

Hemoglobin (Hb) 135 - 175 g/L

WHAT IT MEASURES

The oxygen-carrying capacity of your blood.

WHY IT MAY MATTER

Low levels indicate anemia; high levels can suggest dehydration.

Hematocrit (Ht) 40 - 54%

WHAT IT MEASURES

The percentage of red blood cells in your total blood volume.

WHY IT MAY MATTER

Low levels indicate anemia; high levels suggest an increased risk of thrombosis.

MCV 80 - 100 fL

WHAT IT MEASURES

The average size of your red blood cells.

WHY IT MAY MATTER

Low values indicate microcytic anemia; high values point to macrocytosis.

MCH 27 - 33 pg

WHAT IT MEASURES

The amount of hemoglobin per red blood cell.

WHY IT MAY MATTER

Helps in the classification of different types of anemia.

MCHC 32 - 36 g/dL

WHAT IT MEASURES

The concentration of hemoglobin inside the red blood cells.

WHY IT MAY MATTER

Low levels are indicative of hypochromic anemia.

White Blood Cells (WBC) 4.0 - 10.0 $\times 10^9/L$

WHAT IT MEASURES

The total number of immune cells.

WHY IT MAY MATTER

High levels often mean infection; low levels suggest immunodeficiency.

Neutrophils 2.0 - 7.0 $\times 10^9/L$

WHAT IT MEASURES

A type of white blood cell that fights bacteria.

WHY IT MAY MATTER

Elevated levels typically indicate a bacterial infection.

Lymphocytes 1.0 - 4.0 $\times 10^9/L$

WHAT IT MEASURES

A type of white blood cell responsible for immune response.

WHY IT MAY MATTER

Elevated levels are common during viral infections.

Monocytes 0.2 - 0.8 x10⁹/L

WHAT IT MEASURES

Macrophages in circulation.

WHY IT MAY MATTER

Increased levels occur during infections and inflammation.

Eosinophils 0.0 - 0.5 x10⁹/L

WHAT IT MEASURES

Cells involved in the allergic response.

WHY IT MAY MATTER

Elevated by allergies or parasitic infections.

Basophils 0.0 - 0.2 x10⁹/L

WHAT IT MEASURES

Cells involved in immune and inflammatory reactions.

WHY IT MAY MATTER

Rarely elevated, usually pointing to allergies or chronic diseases.

Platelets (PLT) 150 - 450 x10⁹/L

WHAT IT MEASURES

Cell fragments responsible for blood coagulation.

WHY IT MAY MATTER

Low levels increase bleeding risk; high levels increase thrombosis risk.

02

Iron Status

Iron 11 - 30 µmol/L

WHAT IT MEASURES

The mineral essential for oxygen transport.

WHY IT MAY MATTER

Low levels indicate anemia; high levels can suggest hemochromatosis.

03

Glucose & Metabolic Health

HbA1c < 5.7%

WHAT IT MEASURES

Your average blood sugar levels over the past 2 to 3 months.

WHY IT MAY MATTER

Higher values are indicative of prediabetes or diabetes.

04

Liver

AST < 40 U/L

WHAT IT MEASURES

A liver enzyme associated with liver and muscle damage.

WHY IT MAY MATTER

High levels point to inflammation or tissue damage.

ALT < 41 U/L

WHAT IT MEASURES

An enzyme highly specific to the liver.

WHY IT MAY MATTER

High levels suggest a possible fatty liver or hepatitis.

ALP 44 - 147 IU/L

WHAT IT MEASURES

The activity of the bile ducts and bone turnover.

WHY IT MAY MATTER

Elevated levels indicate potential liver or bone issues.

Albumin 35 - 52 g/L

WHAT IT MEASURES

The main protein found in blood plasma.

WHY IT MAY MATTER

Low levels can indicate chronic disease or poor liver function.

05

Kidney

BUN 7 - 20 mg/dL

WHAT IT MEASURES

The amount of nitrogen from urea in the blood.

WHY IT MAY MATTER

High levels point to kidney problems or dehydration.

Cystatin C 0.6 - 1.0 mg/L

WHAT IT MEASURES

A highly sensitive marker of kidney function.

WHY IT MAY MATTER

Elevated levels indicate reduced kidney filtration.

eGFR > 90 mL/min

WHAT IT MEASURES

The estimated filtration rate of the kidneys.

WHY IT MAY MATTER

A low eGFR is indicative of chronic kidney disease.

06

Lipids / Cardiovascular Health

ApoB < 80 mg/dL optimal

WHAT IT MEASURES

The number of particles carrying "bad" cholesterol.

WHY IT MAY MATTER

Elevated ApoB signifies a greater risk for atherosclerosis.

Lipid Panel (LDL / HDL / TG) LDL < 100, HDL > 40, TG < 150

WHAT IT MEASURES

The profile of LDL, HDL, and Triglycerides.

WHY IT MAY MATTER

A balanced lipid profile is essential for protecting the heart.

07

Thyroid

TSH, fT3, fT4 TSH 0.4-4.0 | fT3 2.5-4.3 | fT4 0.8-1.8

WHAT IT MEASURES

The regulation and function of the thyroid gland.

WHY IT MAY MATTER

Imbalances point to either hypothyroidism or hyperthyroidism.

08

Hormones

Total Testosterone 8 - 29 nmol/L

WHAT IT MEASURES

The primary male sex hormone.

WHY IT MAY MATTER

Low—weak libido and energy; High—often due to physical or pharmacological stimulation.

Free Testosterone 0.2 - 0.6 nmol/L

WHAT IT MEASURES

The biologically active portion of testosterone.

WHY IT MAY MATTER

The most important indicator of actual hormone activity.

SHBG 18 - 54 nmol/L

WHAT IT MEASURES

The binding protein for sex hormones.

WHY IT MAY MATTER

Excess reduces free testosterone; a deficit is linked to metabolic syndrome.

DHT 0.3 - 3.0 ng/mL

WHAT IT MEASURES

A more potent derivative of testosterone.

WHY IT MAY MATTER

High—risk of hair loss; Low—reduced libido and strength.

DHEA-S 3.6 - 11.1 $\mu\text{mol/L}$

WHAT IT MEASURES

The precursor of androgen hormones from the adrenal glands.

WHY IT MAY MATTER

Higher—vitality; Lower—adrenal exhaustion.

LH 1.7 - 8.6 IU/L

WHAT IT MEASURES

Stimulates the testes to produce testosterone.

WHY IT MAY MATTER

Low—pituitary dysfunction; High—testicular dysfunction.

FSH Reference not provided

WHAT IT MEASURES

Stimulates spermatogenesis.

WHY IT MAY MATTER

Abnormal levels indicate possible reduced fertility.

Estradiol (E2) 10 - 40 pg/mL

WHAT IT MEASURES

The main estrogen in men and women.

WHY IT MAY MATTER

Too high—gynecomastia, water retention; Too low—weak libido, bone issues.

Progesterone 0.2 - 1.4 ng/mL

WHAT IT MEASURES

A precursor for testosterone and cortisol.

WHY IT MAY MATTER

Low levels suggest an imbalance in relation to estrogen.

Prolactin 86 - 324 mIU/L

WHAT IT MEASURES

Regulates reproductive functions.

WHY IT MAY MATTER

High—low libido, low testosterone. Note: Must be awake for at least 2 hours before blood draw!

Vitamins & Nutrients

Vitamin D

> 30 ng/mL

WHAT IT MEASURES

Important for bone health, immunity, and hormones.

WHY IT MAY MATTER

Low levels cause weak immunity and lower testosterone.

Inflammation

hsCRP

< 1 mg/L

WHAT IT MEASURES

High-sensitivity C-reactive protein.

WHY IT MAY MATTER

Elevated levels indicate hidden inflammation and a greater risk for a heart attack.

Homocysteine

< 10 µmol/L

WHAT IT MEASURES

An amino acid linked to cardiovascular risk.

WHY IT MAY MATTER

Elevated levels point to a B vitamin deficit and heart risk.

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